

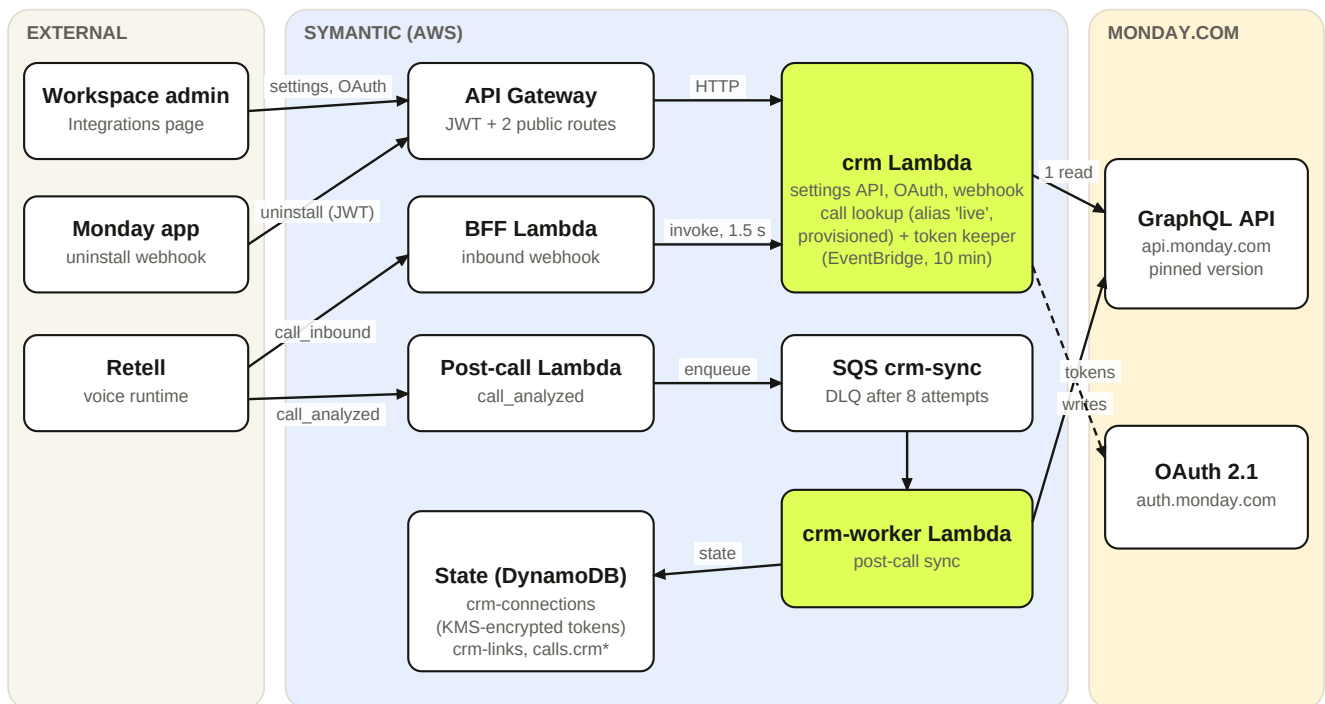
Monday CRM integration

How the AI receptionist uses Monday.com, and why it is built this way. Describes the implementation merged in symantic-agents-infra (PRs #46, #47) and symantic-agents-frontend (PR #71).

What it does. When a call comes in, the receptionist looks the caller up in the business's Monday board and, if they are known, greets them with that context. After every call, a background worker finds or creates the caller's Monday record and adds one note with the summary, outcome, appointment and follow-up, and keeps a few columns the receptionist owns up to date. Each business connects its own Monday account with OAuth and chooses which board and columns are used.

Decision	Choice
Integration path	Direct Monday GraphQL API, authorized per business through a Monday OAuth 2.1 app
Domain boundary	CrmProvider contract; Monday lives only in lambda/crm/monday/
Live call	One bounded read (1.5 s cap, fail-open) on a warm, provisioned Lambda alias; no writes, no token refresh (a 10-minute keeper refreshes tokens ahead of expiry)
After the call	SQS queue -> crm-worker Lambda -> Monday; DLQ after 8 attempts
Inbound events	Only Monday's app-uninstall webhook; no board webhooks, no polling
Duplicates	Per-phone lease, search-before-create, Monday Idempotency-Key, Ref: marker, watermark

1. Architecture



Two new Lambdas share one package (lambda/crm): **crm** serves the settings API, the OAuth callback, the lifecycle webhook and the call-time lookup; **crm-worker** consumes the sync queue. Only these two hold the Monday KMS key and app secret. The BFF can only invoke the lookup and read connection status; the post-call Lambda can only read connection status and enqueue.

2. Layering: CrmProvider -> Monday adapter -> Monday API

Receptionist domain

sync.mjs, lookup.mjs, facts.mjs, activity.mjs, context.mjs - provider-neutral

CrmProvider contract

provider.mjs - CrmContact, CrmFieldPatch, CrmActivity, CrmError

MondayCrmAdapter

monday/adapter.mjs - columns, labels, board scoping; monday/session.mjs - tokens

Monday GraphQL + OAuth clients

monday/graphql.mjs, monday/oauth.mjs - HTTP, errors, idempotency keys

The sync and lookup code never sees a board id, column id, label or GraphQL shape - only these domain types. A HubSpot or Salesforce adapter implements the same six methods and registers in the provider registry; nothing in the receptionist changes.

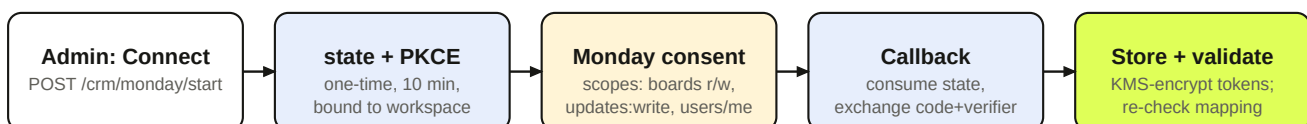
```
findContactByPhone(session, e164)      -> CrmContact | null
findContactByEmail(session, email)     -> CrmContact | null
getContact(session, externalId)        -> CrmContact | null
createLead(session, lead, {idempotencyKey}) -> CrmContact
logCallActivity(session, id, activity, {fields, idempotencyKey})
                                         -> {activityId, fieldsApplied}
findActivityByRef(session, externalId, ref) -> activityId | null
All methods throw CrmError {code, retryable, retryAfterSeconds}
```

3. Why direct API + OAuth app

The receptionist needs one fast, predictable read during a live call and a few exactly-once writes after it, for many independent businesses. That rules the options in or out:

Option	Verdict	Deciding reasons
Direct GraphQL API + OAuth 2.1 app	Chosen	Per-business consent with scopes; revocation and uninstall events; full control of timeouts, batching (note + fields in one call), error codes, idempotency keys and rate-limit handling; no third party in the data path.
Personal API tokens	Rejected	A pasted token carries the user's full access with no scopes, never expires, and its revocation is invisible to us. Poor multi-tenant onboarding.
Monday MCP server	Rejected for this path	A user-scoped wrapper over the same GraphQL API with the same daily cap, plus an extra hop and tool-schema indirection that hides the error codes retries depend on. Letting the voice LLM write to the CRM freely would also let a caller steer CRM changes. Suitable later for a staff-facing 'ask your CRM' feature.
Zapier / Make / n8n	Rejected	Each business must build and maintain its own zap; no call-time lookup; per-task cost; debugging split across vendors; credentials held by a third party.
Board webhooks mirroring Monday	Not needed now	The only Monday-owned data we use (name, status, owner) is read fresh with one targeted query when a call arrives - no polling and no mirror to keep consistent. Revisit if Symantic UI must show CRM status or react to Monday changes.

4. Authentication and tenant configuration

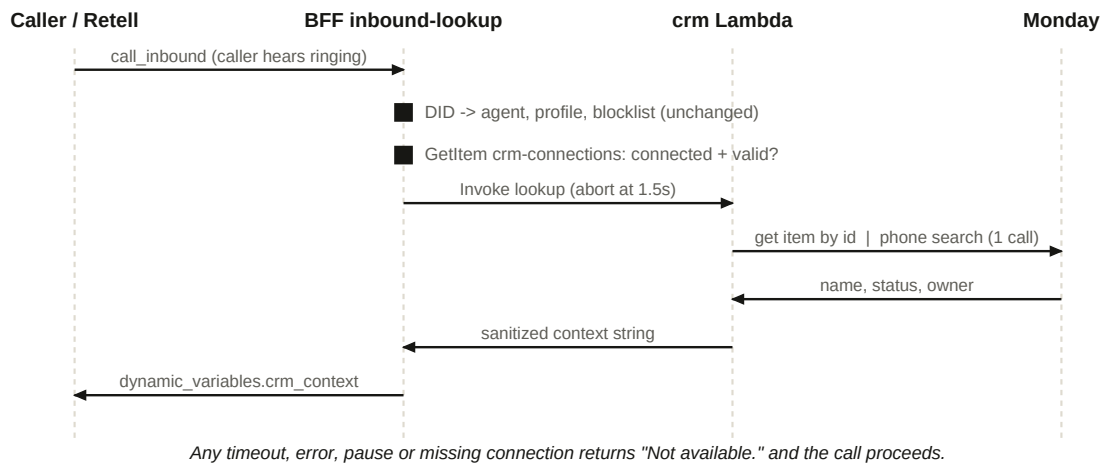


Access token: 1 h, refreshed under a row lock with version compare-and-swap. Refresh token: rotates every use; hard 6-month limit -> admin reconnects.

- Legacy Monday tokens stop working on 2026-10-01; this is built on OAuth 2.1 only (PKCE S256, rotating refresh tokens). Access tokens (1 h) and refresh tokens are KMS-encrypted with a dedicated key; the encryption context binds each ciphertext to its workspace and purpose.

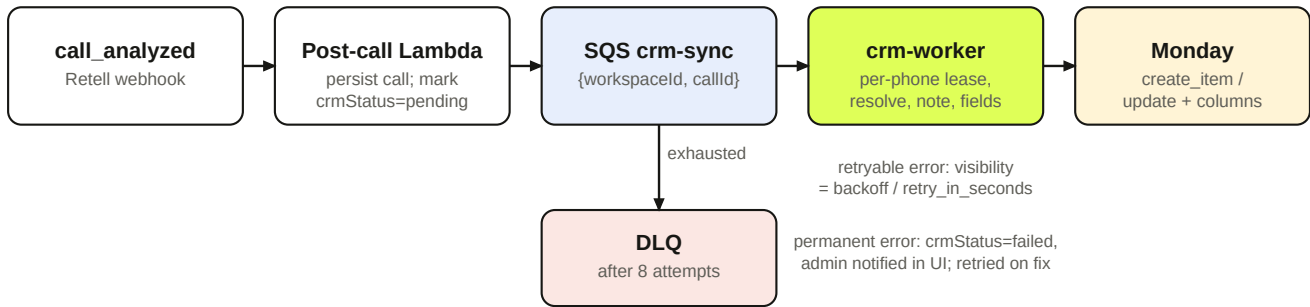
- A token keeper (EventBridge, every 10 minutes) refreshes any access token with under 25 minutes left, so the call-time lookup never refreshes - and a revoked grant is found by the keeper, not by a caller. It refreshes five accounts at a time within a 10-second budget; one account's failure never stops the rest.
- Refresh is serialized across Lambda containers by a short lock on the connection row and committed with a tokenVersion compare-and-swap, so a rotated refresh token is never lost to a race. A rejected refresh or a second 401 marks the connection *reauth_required*; the UI shows Reconnect.
- Monday caps a grant at six months from consent. The UI shows 'Reconnect by' and prompts 14 days ahead. Reconnecting keeps the mapping, re-validates it, and re-queues calls that failed meanwhile.
- Field mapping: the admin picks a board (only boards with a Phone column), then which columns hold phone (required), email, status, owner, last call, outcome, follow-up date, next appointment and source, plus the new-lead and follow-up status labels and a default owner. It is validated live against Monday; column types are taken from Monday, never from the browser.
- Disconnect revokes the grant at Monday (best effort) and deletes our tokens. Monday's app-uninstall webhook (HS256 JWT, client secret, account and app claims checked) disconnects every workspace on that account.

5. During the call: caller lookup



- The BFF invokes the lookup through the crm Lambda's *live* alias, which keeps a provisioned, pre-initialized instance (SDK clients built at init) - no cold start while the caller hears ringing. The settings API and the keeper use the unqualified function and never take that capacity.
- Started only after the call is accepted (a blocked or rejected caller costs no Monday call), in parallel with the overage check. Hard budget: 1.5 s including the Lambda invoke; the invoke is aborted at the deadline. Retell allows 10 s here while the caller hears ringing.
- Workspaces without a usable connection skip the invoke after a single DynamoDB read.
- Known callers are fetched by record id; unknown ones cost one phone search. The answer is written to crm-links so the post-call sync can skip a repeat search.
- CRM text is untrusted: markup, template braces and control characters are stripped, fields capped at 60 characters. The prompt's CALLER RECORD section tells the agent to confirm the name before using it and never to disclose what the record says. A Retell LLM default makes test and web calls read 'Not available.'

6. After the call: asynchronous sync



One job per call, keyed by callId. Test calls, spam and callers with no number are never enqueued. The worker:

- takes a lease on the caller's phone (crm-links) so two calls from one new number cannot create two leads;
- resolves the record: stored link -> phone search -> email search -> create (with a 'creating' marker first, so a crash after Monday commits is followed by a search, not a second create);
- posts one note and, for existing records, updates the columns we own **in the same GraphQL request**;
- advances a per-phone watermark so an older call processed late never overwrites a newer call's fields.

Caller	Monday API calls per call	Requests
Returning (known at call time)	2	get item (lookup) + note & fields (sync)
New	3	search (lookup) + create_item + note (sync)
New, lookup skipped/failed	4	+1 search in the worker

Monday's daily cap is per account and shared with the business's other integrations (1,000 on Standard, 10,000 on Pro). Worker concurrency is capped at 5, below Monday's lowest concurrency limit (40).

7. Idempotency and duplicate protection

Risk	Protection
Retell repeats call_analyzed / SQS redelivers	calls.crmStatus=synced short-circuits; re-arming a synced call is a conditional write that fails
Two calls from the same new number at once	Per-phone lease (conditional write, 2 min) - the second job backs off 15 s
create_item committed but response lost	'creating' marker forces a search on retry; plus Monday Idempotency-Key (30 min replay window)
Note committed but response lost	Idempotency-Key within 30 min; after that, the item's recent updates are searched for the note's 'Ref: <callId>' line before posting
Request changed between attempts (mapping fixed)	Idempotency keys are call + intent + hash of the request body, so a changed request is new, an identical retry is a replay
Out-of-order calls	lastAppliedEndedAt watermark on the phone link; notes are always recorded
Record deleted in Monday	not_found on write -> link cleared -> resolved again once (find or create)

8. Source of truth and field mapping

Data	Owner	Sync
Call, transcript, recording, summary, intent, outcome	Symantic	Summary and outcome into the note; transcript and recording never leave Symantic (the note links back)
Appointment	Calendar provider (via Symantic)	One-way: 'Next appointment' date; cleared only if the cancelled appointment is the one we wrote
Lead/contact record, name, email	Monday	Written only when we create the record

Data	Owner	Sync
Lead status	Monday	Set on creation; set to the follow-up label only if the admin chose one
Owner / salesperson	Monday	Default owner on creation only; never changed afterwards
Last call, call outcome, follow-up date	Symantic	One-way, newest call wins
Caller phone -> Monday item	Symantic (crm-links)	Reference only

Nothing syncs in both directions, so there is no loop to break.

Receptionist field	Monday column type	Written when
Phone (required)	phone	new lead; also the match key
Email	email	new lead; secondary match key
Lead status	status	new lead ('new lead' label); follow-up label if configured
Owner	people	new lead (default owner)
Last call / Follow-up date / Next appointment	date	every call (newest wins)
Call outcome / Lead source	text or long text	every call / new lead
Call note	item update	every call, exactly once

9. Failure handling

Condition	During the call	Post-call sync
Monday slow / timeout / 5xx	No context; call continues	Retry: 30 s, 60 s, 2 min ... 15 min, then DLQ
Rate limited (429, complexity)	No context	Retry after Monday's retry_in_seconds
Daily API cap	No context; lookups paused	Connection paused until 00:05 UTC; messages wait, then catch up
Token near expiry	Never happens: the keeper refreshes 25 min ahead	Refreshed; one retry on 401
Grant revoked / 6-month limit	No context	reauth_required; call marked failed; re-queued on reconnect
Board or column deleted, label missing	No context	Mapping marked invalid, call failed (no pointless retries); re-queued when the admin fixes the mapping
Unexpected bug / AWS error	No context	Bounded retries, then DLQ + alarm

10. Security and tenant isolation

- Every read and write is keyed by the workspaceId from the caller's verified Cognito identity (or from a row we wrote) - never from Monday or the request body. Settings routes require the JWT authorizer; mutating routes require company-admin or super-admin.
- Monday scopes are not board-granular, so the adapter writes only to the mapped board and ignores records from any other board.
- Tokens are stored only as KMS ciphertext, never logged, never returned to the browser; the accountId index projects keys and state only. OAuth state is single-use, 10-minute, bound to workspace and redirect URI; returnTo cannot leave the app.
- Logs carry workspace and call ids, error codes and masked numbers (***0198) - no names, summaries or tokens.
- IAM is per function: BFF may invoke the lookup and read connection status; post-call may read status and enqueue; only crm/crm-worker can use the KMS key and the Monday secret.

11. Observability

Signal	Source
ApiLatency, ApiCall (by operation and outcome)	Symantic/CRM metrics (EMF logs)
InboundLookupLatency (by outcome), Lookup, LookupLatency	BFF and crm Lambda
SyncSucceeded / SyncFailed / SyncRetried / DeadLettered / SyncDuration / QueueAge	crm-worker
RateLimited, TokenRefresh, TokenFailure, KeeperRefreshed, KeeperFailed, OAuth, Webhook, LeaseBusy, StaleLink	crm and crm-worker
Alarms: DLQ not empty; queue older than 1 h; worker errors; >=10 lookup timeouts / 15 min; >=5 mapping failures / h	CloudWatch (optional SNS via crm_alarm_topic_arn)

12. Operations

One-time setup (per environment). Register a Monday app in the Developer Center with the 'New OAuth flow' enabled; redirect URI = terraform output monday_oauth_redirect_uri; scopes me:read, account:read, boards:read, boards:write, updates:write, users:read; lifecycle webhook URL = terraform output monday_lifecycle_webhook_url. Store the credentials:

```
aws secretsmanager put-secret-value --secret-id symantic/dev/monday-oauth \
  --secret-string '{"clientId":"...", "clientSecret":"...", "signingSecret":"...", "appId":"..."}'
```

- **DLQ redrive** after fixing the cause: SQS console 'Start DLQ redrive' (or start-message-move-task). Replays are safe - sync is idempotent.
- **Admin-fixable failures** (reconnect, mapping) retry themselves when fixed; 'Retry failed calls' on the Integrations page re-queues the last 7 days.
- **Capacity knobs:** crm_provisioned_concurrency (default 1, about \$3/month) sizes the warm lookup instances; the keeper schedule is the crm-token-keeper EventBridge rule.
- **Monday API version** is pinned (2026-07, variable monday_api_version). Move it forward deliberately when Monday announces its deprecation, and rerun the suites.
- **Existing agents** receive the prompt's CALLER RECORD section the next time they are saved (as with every prompt change); until then their calls still sync after the call, only without in-call context.

Known limitations. Calls log as item updates, not Monday CRM 'Emails & Activities' timeline entries (that API has no idempotency key and needs a per-account activity type). Changes made in Monday are read at call time, not pushed to Symantic. Numbers on +1 are classified US unless the area code is Canadian; other NANP countries may be rejected by Monday's phone validation (the call fails as invalid_value, visibly).

13. Verification

Deployed to dev (agents.symantic.ai) on 2026-09-25 and verified with a synthetic workspace and fictional number; Monday calls were real, against a deliberately invalid token, because the Monday app credentials are not registered yet. A successful Monday write is covered by the flow tests only.

Check	Result	Covers
CRM Lambda unit + flow tests	89 passing	Adapter, OAuth, sessions, keeper, sync, lookup, worker, webhook, isolation
Post-call / BFF / other Lambda suites	29 / 403 / 142 passing	CRM enqueue + inbound lookup; full regression
Frontend	625 passing; lint, tsc, build clean	Connect, callback errors, mapping editor, states, retry
DynamoDB integration (real tables)	6 / 6	CAS, refresh lock, leases, watermark, null-remove, GSI projection
Deployed: inbound call, no CRM	200, unchanged	crm_context 'Not available.'; no invoke
Deployed: Monday rejecting	all calls answered	Fail-open; connection flagged reauth_required
Deployed: post-call -> worker	8 s queue-to-done	Call row written; sync failed cleanly (reauth); 1 row per call
Deployed: daily-cap pause	resumed after 65 s	Message deferred for the 60 s pause, then processed
Deployed: settings API	as designed	503 not configured, 403 member, 401 anon, forged state rejected
Deployed: keeper / DLQ	flagged in 1 run / DLQ after 8, redrive OK	Keeper flagged a dead grant before any call; poison message dead-lettered after 8 attempts (DLQ alarm fired), then replayed once after the fix
Deployed: tools, post-call regression	OK	Existing Retell tool endpoint and post-call ingest unchanged

Live-call latency

Path (measured end to end from the client, 12 samples)	p50	p90
Inbound webhook, no CRM connected (baseline)	317 ms	480 ms
CRM connected: status read + warm 'live' invoke	345 ms (+28 ms)	556 ms
Monday round trip from Lambda (API logs)	121-199 ms warm	586-859 ms new TLS
Healthy lookup, derived: overhead + one Monday read	~+0.2 s	~+0.9 s; hard cap 1.5 s

Before the token keeper and the provisioned alias, a lookup that had to refresh the token took 1.4-2.8 s and ran past its budget - found by this measurement and fixed in infra PR #47.